

Module 3 In-class Coding Practice - Inheritance

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Complete the following skeleton codes:

Car Skeleton:

```
class Car {  
    private String make;  
    private String model;  
  
    // Constructor  
    public Car(String make, String model) {  
        // Initialize make and model  
    }  
  
    // Start method  
    public void start() {  
        // Implement start method  
    }  
  
    // Drive method  
    public void drive() {  
        // Implement drive method  
    }  
  
    public static void main(String[] args) {  
        // Create a Car object and test its methods  
    }  
}
```

Book Skeleton:

```
class Book {  
    private String title;  
    private String author;  
  
    // Constructor  
    public Book(String title, String author) {  
        // Initialize title and author  
    }  
  
    // DisplayInfo method  
    public void displayInfo() {  
        // Implement displayInfo method  
    }  
  
    public static void main(String[] args) {  
        // Create a Book object and test its methods  
    }  
}
```

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Person Skeleton:

```
class Person {  
    private String name;  
    private int age;  
  
    // Constructor  
    public Person(String name, int age) {  
        // Initialize name and age  
    }  
  
    // Introduce method  
    public void introduce() {  
        // Implement introduce method  
    }  
  
    public static void main(String[] args) {  
        // Create a Person object and test its methods  
    }  
}
```

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Animal Skeleton:

```
class Animal {  
    private String species;  
  
    // Constructor  
    public Animal(String species) {  
        // Initialize species  
    }  
  
    // MakeSound method  
    public void makeSound() {  
        // Implement makeSound method  
    }  
  
    public static void main(String[] args) {  
        // Create an Animal object and test its methods  
    }  
}
```

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Plant Skeleton:

```
class Plant {  
    private String name;  
  
    // Constructor  
    public Plant(String name) {  
        // Initialize name  
    }  
  
    // Grow method  
    public void grow() {  
        // Implement grow method  
    }  
  
    public static void main(String[] args) {  
        // Create a Plant object and test its methods  
    }  
}
```

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These skeleton codes provide the basic structure of the `Car`, `Book`, `Person`, `Animal`, and `Plant` classes. You can fill in the constructor and method implementations as needed for your specific application.

Grading Rubric

Criteria	Points
Class Definition	
- Class name matches the entity it represents (e.g., Car for a car entity)	5
Constructor	
- Constructor initializes instance variables correctly	10
Class Methods	
- Class methods are defined appropriately (e.g., start, drive, displayInfo)	10
- Methods are implemented correctly with appropriate logic	15
Code Structure	
- Code is well-structured with appropriate indentation and formatting	10
- Proper use of access modifiers (private, public)	10
- Main method tests class methods effectively	10
Documentation	
- Class, constructor, and method comments provide clear descriptions	10
- Code follows naming conventions for variables, methods, and classes	5
Total Points	100

Points 100

Submitting a file upload

Due	For	Available from	Until
Sep 24	Everyone	-	Oct 22 at 11:59pm